Study of Unsupervised Error Logs Clusterization Methods

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The Key Objectives of Error Logs Clusterization

- Computing infrastructures have grown significantly during the last decade. Heterogeneity, distribution, complexity, friendly and flexible interfaces have become a subject to many dynamic failures.

- Error logs are a fruitful source of information:
  - Search for new (previously unknown) error messages
  - Categorize error messages sharing similar meaning/structure

- Diagnosing failures is a challenging problem as it requires the analysis of large amounts of textual log data without well-defined structure

- Existing approach: Human experts interpret detected log errors and create the rules (patterns of regexp) manually
  - Time-consuming
  - Doesn't allow to identify previously unknown error conditions

✓ A possible solution to this issue is to cluster log errors using methods of text clusterization.

Requirements for Error Logs Clusterization Approach

- Performance, suitable for the real-time applications
- The number of clusters is unknown in advance
- Each outlier (previously unknown error message) should be treated as a separate cluster
- Minimization of human intervention in the process of clusterization, relying on automatic detection of clusterization parameters, learned from data

Error Logs Test Dataset

- ATLAS BigPanDA jobs archive
  - ~5K exeerrordiag unique messages
- Some examples:
  - Fatal error in athena logfile: "Logfile error in log.SUSY16Merge: "ToolSvc.FileMetaDataTool FATAL Failed to call beginInputFile()""
  - Missing AthenaMP outputs file athenaMP-outputs-EVNTtoHITS-sim (probably athena crashed)
  - LRMS error: (91) Job failed
  - Non-zero return code from OverlayPool (2); Logfile error in log.OverlayPool: "IncludeError: include file NULL can not be found"
  - Input file argument MC14.Pileup.RDO.pool.root globbed to NO input files - probably the file(s) are missing

1) Strings
2) Vector Space Model
3) Clusterization

Existing approaches

Strings Clusterization Pipeline without Preprocessing

Array of error messages

Levenshtein distances

finding the number of edits **(insert | delete | replace)** which will transform one string to another

String 1	String 2	Levenshtein distance
aaa	aaa	0
aaa	aba	1
aaa	abcdefg	6

Distance Matrix (5000 x 5000)

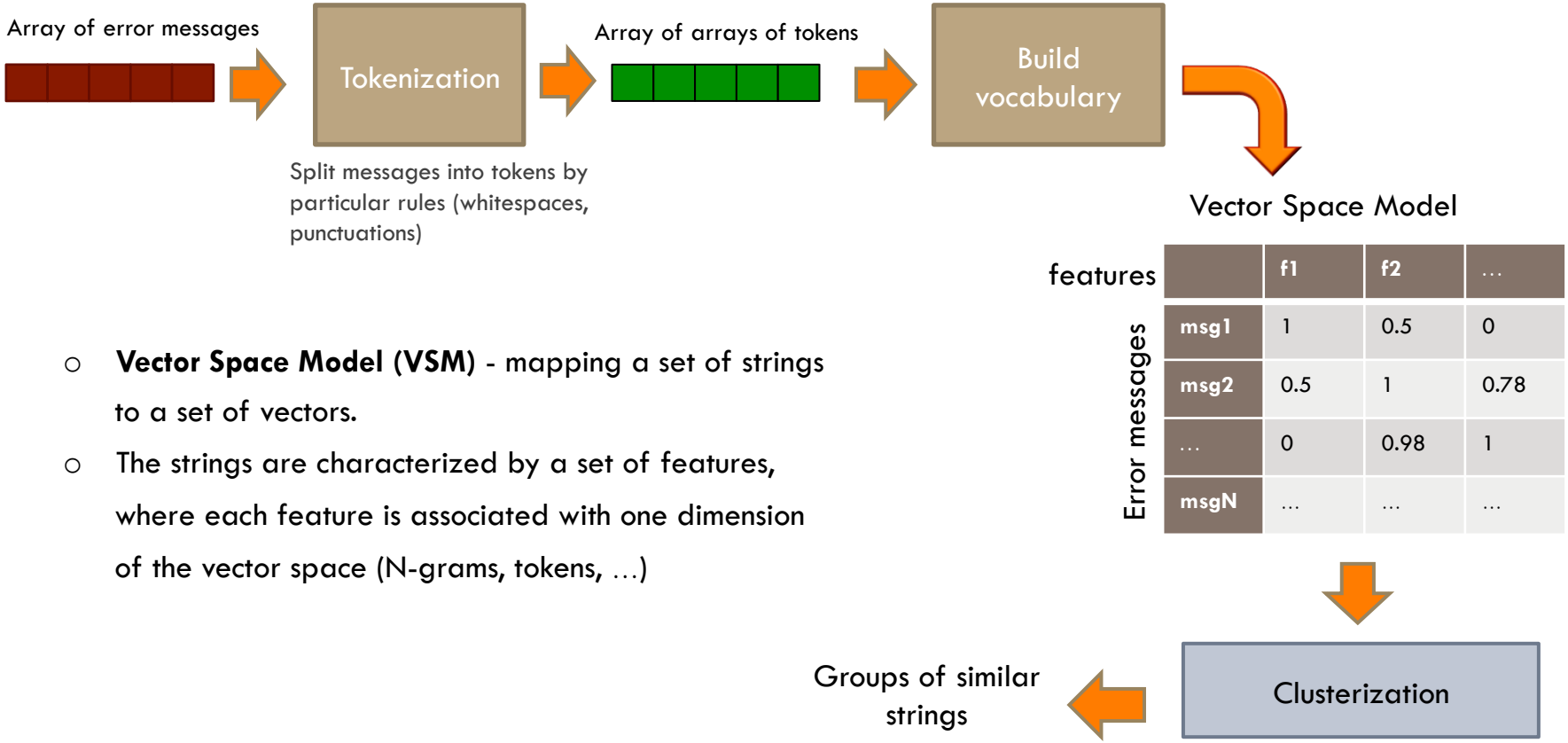
	msg1	msg2	msg3
msg1	0	10	50
msg2	10	0	1
msg3	50	1	0

Clusterization

Groups of similar strings and outliers

- Advantages:
  - No need for text preprocessing
- Disadvantages:
  - Levenshtein distances are calculated very slowly
  - Levenshtein distances don't take into account strings structure
  - Clusterization results suffer of noise

Strings Clusterization Pipeline with Preprocessing



- **Vector Space Model (VSM)** - mapping a set of strings to a set of vectors.
- The strings are characterized by a set of features, where each feature is associated with one dimension of the vector space (N-grams, tokens, …)

Vector Space Model = TF-IDF + Cosine Similarity

Array of error messages

Tokenization

Array of arrays of tokens

Build vocabulary

Vectorization

TF-IDF (Term Frequency – Inverse Term Frequency) Matrix **(5 000 x 10 000)**

	token1	token2	…
msg1	0.98	0.33	…
msg2	0.65	0.78	…
msg3	0.34	0.12	…
…	0.90	1.00	…
…	0.08	0.09	…

VSM = Cosine Similarity Matrix **(5 000 x 5 000)**

	msg1	msg2	…
msg1	1.00	0.78	…
msg2	0.78	1.00	…
msg3	0.00	0.01	…

The cosine similarity between 2 messages is a function of the angle between their vectors in the VSM.

Clusterization

Groups of similar strings

- Advantages:
  - Good at performance as log messages strings are not very long
- Disadvantages:

  TF-IDF limitations
  - ignores tokens order and structure of messages
  - In most cases it generates large sparse matrix (due to large vocabulary)

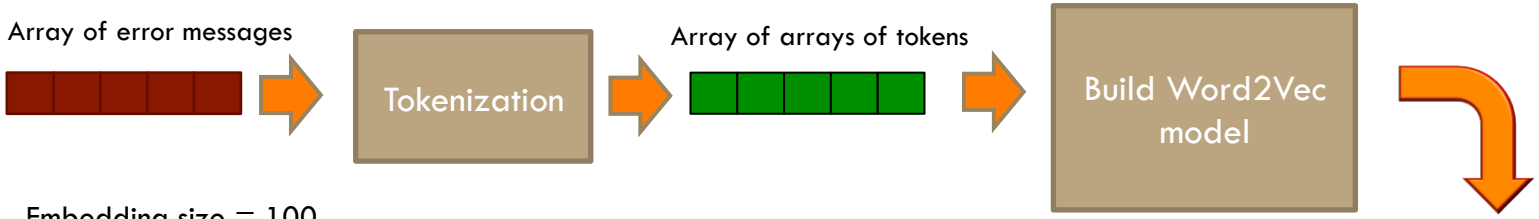
The resulting VSM are not satisfied for error logs, where structure of messages in crucial.

Vector Space Model = Word2Vec Embeddings

- Word2Vec was created and published in 2013 at Google.
- The idea of Word2Vec – meaning of a word can be inferred by the company it keeps

  - CBOW (Continuous Bag-of-Words) – predicts the current word w, given the neighboring words in the window
  - Skip-Gram – predicts the neighboring words given the current word
- Python library Gensim Word2Vec
  - **model = Word2Vec(sentences, size=200, window=5, min_count=1)**
    - **size** (of embedding) – context dimensionality
    - **window** – the max distance between the target word and its neighbors
    - **min_count** – minimum frequency of count of words

Conversion of Word2Vec Model to Messages Matrix



Embedding size = 100
X ~ 10 000 (size of vocabulary)
N = 5 000 (number of error messages)

Messages Vector Model (**5 000 x 200**)

	f0	f1	…	f100
msg1	0.98	0.33	…	…
msg2	0.65	0.78	…	…
…	…	…	…	…
msgN	…	…	…	…

Word2Vec Model (**10 000 * 200**)

	f1	f2	…	f100
token1	0.98	0.33	…	…
token2	0.65	0.78	…	…
…	…	…	…	…
tokenX	…	…	…	…

Clusterization

Groups of similar strings

- **Advantages:**
  - Word2Vec retains the semantic meaning of different words in a document
  - The context information is not lost
  - The size of embedding vector is very small

Clusterization

DBSCAN

DBSCAN Clusterization

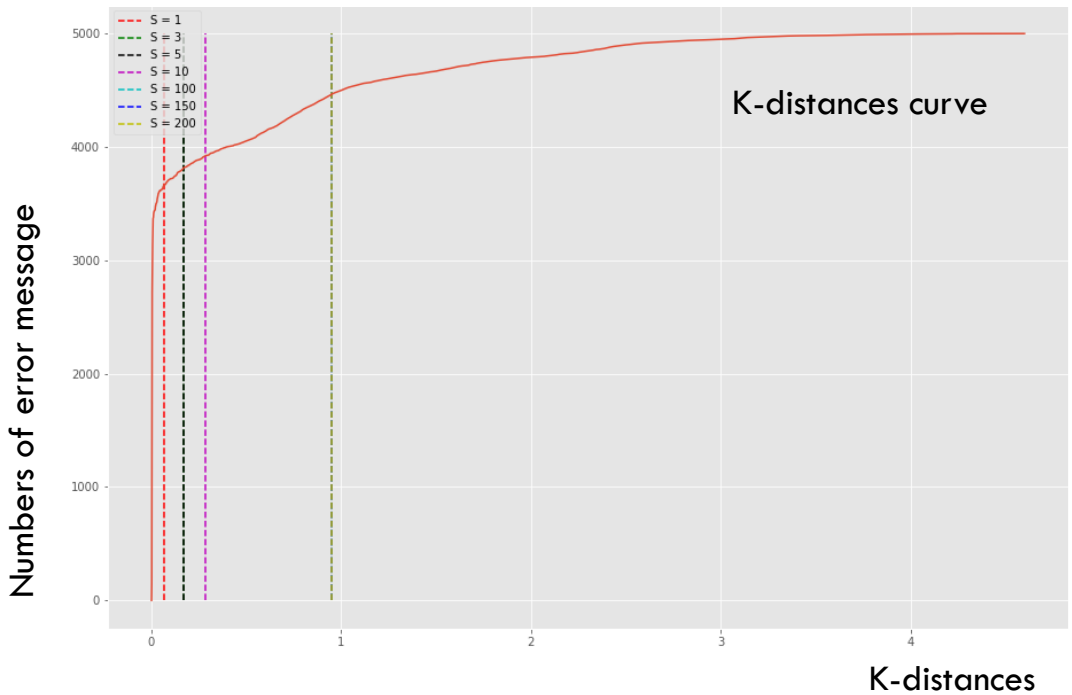
- DBSCAN does not require to specify the exact amount of clusters in advance
- DBSCAN can find any shape of clusters (not only circular)
- DBSCAN can identify outliers
- Requires 2 parameters:
  - **eps** – the maximum distance between two samples for one to be considered as in the neighborhood of the other
  - **min_samples** – the number of samples in a neighborhood for a point to be considered as a core point

What are the optimal min_samples and epsilon for DBSCAN ?

- **min_samples = 1**
  - It guarantees that each outlier will be put in separate cluster
  - Otherwise, if min_samples > 1, all outliers are put in a garbage cluster "-1"
- **epsilon**
  - Calculate the average of the distances of every point to its k-neighbors
    - $k = \sqrt{n}$,where *n* is the number of samples in dataset
  - Sort these k-distances in an ascending order
  - Find a knee on a plot where a sharp change occurs along the k-distance curve
- But this method is not automatic as it requires user interpretation of k-distance curve.

Automatic Detection of Knee for k-distance Curve with kneed/python

- Kneed is a python library allowing to detect knees for curves automatically
- It shows knees graphically and returns knee points.



[0.06899074925405664,
0.17418354191659477,
0.17418354191659477,
0.2850125475866659,
0.9484519887921964,
0.9484519887921964,
0.9484519887921964]

Epsilon (DBSCAN)

Clusterization Results

- Array of 5000 error log messages
- Word2Vec Model with size=200 and window=5
- Word2Vec converted to messages vectors
- DBSCAN clusterization with automatically detected epsilon value
- DBSCAN Wall-time: ~3
- Results: 73 clusters
  - 31 clusters (size=1) are detected as anomaly error messages
  - to re-check similarity between all messages in each cluster, Levenshtein distances were used

cluster_name	cluster_size	first_entry	mean_length	mean_similarity	std_lengt	std_similarity
1	1669	Failed in data staging: Failed to prepare dest...	498.361893	95.956860	15.700346	2.386123
0	689	Non-zero return code from AODtoDAOD (65); Long...	108.994194	95.653120	0.961449	3.662367
5	683	Non-zero return code from RDOtoRDOTrigger (65)...	201.647145	83.326501	9.584014	9.568720
2	402	Fatal error in athena logfile: "G4 exception a...	95.099502	96.609453	0.307917	0.795380
4	381	File data17_13TeV.00339957.physics_Main.merge....	71.979003	59.860892	12.590915	12.156396
6	336	Non-zero return code from AODMerge (64); Logfi...	168.666667	63.821429	37.874919	13.119802
12	114	Failed in data staging: Failed to prepare dest...	478.736842	81.850877	49.960525	4.771287
3	104	Non-zero return code from HITtoRDO (8); Logfil...	155.750000	93.730769	10.343376	9.173866
16	92	Non-zero return code from AODtoDAOD (65); Logf...	208.021739	78.010870	25.076138	9.811962
8	91	Failed in data staging: Failed to prepare dest...	480.186813	88.637363	36.683909	7.287000
11	40	AODtoDAOD got a SIGBUS signal (exit code 135);...	177.675000	67.225000	26.487915	8.891043
29	39	Failed in data staging: Could not resolve any ...	222.487179	98.179487	3.042435	2.393762
9	34	RDOtoRDOTrigger got a SIGABRT signal (exit cod...	116.470588	85.882353	4.520896	7.449892
19	33	HITtoRDO got a SIGKILL signal (exit code 137)	47.666667	79.787879	4.511559	7.773181
10	32	Fatal error in athena logfile: "Logfile error ...	170.218750	57.031250	31.323221	10.899791
20	29	Non-zero return code from AODMerge (1)	42.758621	82.724138	6.271564	6.675630
37	26	Fatal error in athena logfile: "Long ERROR mes...	100.461538	97.346154	0.760567	1.412581
21	23	Fatal error in athena logfile: "Logfile error ...	157.521739	97.565217	0.790257	1.273010
14	20	Failed in data staging: Failed checking source...	448.900000	82.950000	34.321851	4.828370
22	18	Input file argument ${TRN_OUTPUT0/L} globbed t...	128.944444	76.777778	44.380006	12.374336
34	12	No events to process: 7000 (skipEvents) >= 500...	68.000000	98.166667	0.000000	1.267304
25	10	LRMS error: (271) Job timeout	32.500000	79.700000	8.195527	14.575855
24	10	Failed in data staging: Failed writing to dest...	301.600000	76.300000	104.565981	12.824023

63	1	User work directory too large	29.000000	100.000000	0.000000	0.000000
64	1	Failed to convert substep value %JOBNUMBER to int	49.000000	100.000000	0.000000	0.000000
67	1	Put error: File copy timed out	30.000000	100.000000	0.000000	0.000000
61	1	LRMS error: (1) Job failed|Failed in data stag...	68.000000	100.000000	0.000000	0.000000
55	1	TRF_SVRINIT | "Caught signal 11(Segmentation f...	250.000000	100.000000	0.000000	0.000000
36	1	Fatal error in athena logfile: "Logfile error ...	122.000000	100.000000	0.000000	0.000000
53	1	AODtoDAOD got a SIGSEGV signal (exit code 139)...	125.000000	100.000000	0.000000	0.000000
13	1	Fatal error in athena logfile: "Logfile error ...	201.000000	100.000000	0.000000	0.000000
15	1	Plugin at state FINISHING failed: TIMEOUT	41.000000	100.000000	0.000000	0.000000
17	1	TRF_UNKNOWN | "Unable to build inputFileSummar...	116.000000	100.000000	0.000000	0.000000
28	1	HITtoRDO got a SIGABRT signal (exit code 134);...	160.000000	100.000000	0.000000	0.000000
35	1	Get error: Input file staging timed out	39.000000	100.000000	0.000000	0.000000
71	1	OK	2.000000	100.000000	0.000000	0.000000
38	1	Put error: Error in copying the file from job ...	64.000000	100.000000	0.000000	0.000000
40	1	Get error: Staging input file failed	36.000000	100.000000	0.000000	0.000000
44	1	Failed in data staging: Checksum mismatch	41.000000	100.000000	0.000000	0.000000
45	1	Failed in data staging: Failed checking source...	500.000000	100.000000	0.000000	0.000000
46	1	File test.EVNT.pool.root might be missing	41.000000	100.000000	0.000000	0.000000
48	1	Error reading user generated output file list	45.000000	100.000000	0.000000	0.000000
49	1	LRMS error: (91) Job missing from SLURM, exitc...	82.000000	100.000000	0.000000	0.000000
50	1	LRMS error: (-1) Job finished with unknown exi...	52.000000	100.000000	0.000000	0.000000
54	1	LRMS error: (-1) Job missing from SLURM	39.000000	100.000000	0.000000	0.000000
72	1	Event count check for DAOD_FTAG2 to DAOD_FTAG2...	256.000000	100.000000	0.000000	0.000000

Cluster 1

['Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/7a/a7/HITS.12906983._043065.pool.root.1:checksumtype=adler32:checksumvalue=4d07cfad: Failed to prepare destination: Temporary service error: Error looking up space tokens matching description (Proxy expired); Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/5e/a5/log.129',

'Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/de/fd/HITS.12906983._039129.pool.root.1:checksumtype=adler32:checksumvalue=b14dece4: Failed to prepare destination: Temporary service error: Error looking up space tokens matching description (Proxy expired); Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/e2/30/log.129',

'Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/ee/fb/HITS.12906983._043921.pool.root.1:checksumtype=adler32:checksumvalue=e609c1b8: Failed to prepare destination: Temporary service error: Error looking up space tokens matching description (Proxy expired); Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/f7/07/log.129',

Cluster 12

['Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/90/9e/log.12790254._060284.job.log.tgz.1:checksumtype=adler32:checksumvalue=ea021253: Failed to prepare destination: File exists: File requests have failed.: The requested SURL is locked by another upload.; Failed in data staging: Checksum mismatch',

'Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/53/eb/log.12906983._044637.job.log.tgz.1:checksumtype=adler32:checksumvalue=00843189: Failed to prepare destination: File exists: File requests have failed.: The requested SURL is locked by another upload.; Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/08/49/HITS.1290',

'Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/8d/28/log.12866455._019450.job.log.tgz.1:checksumtype=adler32:checksumvalue=6071d60b: Failed to prepare destination: File exists: File requests have failed.: The requested SURL is locked by another upload.; Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasdatadisk/rucio/mc16_13TeV/7a/d5/HITS.1286',

Cluster 8

['Failed in data staging: Failed checking source replica file:/boinc-buffer/boinc/session/19HKDm8JtqrnSu7Ccp2YYBZmABFKDmABFKDmG7KKDmABFKDmqrtVBn/log.12906983._019641.job.log.tgz.1: Failed to obtain information about file: No such file or directory: Failed to stat file /boinc-buffer/boinc/session/19HKDm8JtqrnSu7Ccp2YYBZmABFKDmABFKDmG7KKDmABFKDmqrtVBn/log.12906983._019641.job.log.tgz.1; Failed in data staging: Failed checking source replica file:/boinc-buffer/boinc/session/19HKDm8JtqrnSu7Ccp2YYBZmAB',

'Failed in data staging: Failed checking source replica file:/boinc-buffer/boinc/session/sZNMDmOQwqrnSu7Ccp2YYBZmABFKDmABFKDmVLNKDmABFKDmQFWygm/log.12906983._020859.job.log.tgz.1: Failed to obtain information about file: No such file or directory: Failed to stat file /boinc-buffer/boinc/session/sZNMDmOQwqrnSu7Ccp2YYBZmABFKDmABFKDmVLNKDmABFKDmQFWygm/log.12906983._020859.job.log.tgz.1; Failed in data staging: Checksum mismatch',

'Failed in data staging: Failed checking source replica file:/boinc-buffer/boinc/session/dMmKDmJverrnSu7Ccp2YYBZmABFKDmABFKDmdCLKDmABFKDm5B5k2m/log.12906983._037230.job.log.tgz.1: Failed to obtain information about file: No such file or directory: Failed to stat file /boinc-buffer/boinc/session/dMmKDmJverrnSu7Ccp2YYBZmABFKDmABFKDmdCLKDmABFKDm5B5k2m/log.12906983._037230.job.log.tgz.1; Failed in data staging: Checksum mismatch',

Cluster 45 - outlier

['Failed in data staging: Failed checking source replica file:/boinc-buffer/boinc/session/bw1NDmlfHqrnSu7Ccp2YYBZmABFKDmABFKDmGZJKDmABFKDmezSXPo/log.12906983._005962.job.log.tgz.1: Failed to obtain information about file: No such file or directory: Failed to stat file /boinc-buffer/boinc/session/bw1NDmlfHqrnSu7Ccp2YYBZmABFKDmABFKDmGZJKDmABFKDmezSXPo/log.12906983._005962.job.log.tgz.1; Failed in data staging: Failed to prepare destination srm://srm.ndgf.org:8443/srm/managerv2?SFN=/atlas/disk/atlasd']

Cluster 29

['Failed in data staging: Could not resolve any source replicas for rucio://rucio-lb-prod.cern.ch/replicas/mc16_13TeV/HITS.11364157._026091.pool.root.1: Resolving of index service for source failed: No such file or directory',

'Failed in data staging: Could not resolve any source replicas for rucio://rucio-lb-prod.cern.ch/replicas/mc16_13TeV/HITS.11364157._026043.pool.root.1: Resolving of index service for source failed: No such file or directory',

'Failed in data staging: Could not resolve any source replicas for rucio://rucio-lb-prod.cern.ch/replicas/mc16_13TeV/HITS.11364157._009023.pool.root.1: Resolving of index service for source failed: No such file or directory',

'Failed in data staging: Could not resolve any source replicas for rucio://rucio-lb-prod.cern.ch/replicas/panda/panda.1217085143.810334.lib._12830150.12004349677.lib.tgz: Resolving of index service for source failed: No such file or directory',

Cluster 24

["Failed in data staging: Failed writing to destination: file:/ceph/grid/cache/data/8e/e7c79005bd5287c684a33f3f7bfcf71d997434 : Can't write to destination: File exists: Failed to create file /ceph/grid/cache/data/8e/e7c79005bd5287c684a33f3f7bfcf71d997434",

"Failed in data staging: Failed writing to destination: file:/ceph/grid/cache/data/9b/fecc21c035f04eedaa5d470251161d4149241f : Can't write to destination: File exists: Failed to create file /ceph/grid/cache/data/9b/fecc21c035f04eedaa5d470251161d4149241f; Failed in data staging: Failed writing to destination: file:/ceph/grid/cache/data/8c/70223081e536cc6ae39464d5f96b7ed5dbba52 : Can't write to destination: File exists: Failed to create file /ceph/grid/cache/data/8c/70223081e536cc6ae39464d5f96b7ed5",

"Failed in data staging: Failed writing to destination: file:/ceph/grid/cache/data/78/9b5fe022c4c88e93dc86cceb26086b654596c1 : Can't write to destination: File exists: Failed to create file /ceph/grid/cache/data/78/9b5fe022c4c88e93dc86cceb26086b654596c1",

Cluster 7

["Failed in data staging: Failed writing to destination: file:/disk/session/aurora-gc3-session1/VAzKDm8YHqrnjVCY5mifW1amABFKDmABFKDmB9IKDmABFKDmP6ybOm/DAOD_EGAM8.12607600._000963.pool.root.1.panda.um_010 : Can't write to destination: No space left on device: Failed to create file /disk/session/aurora-gc3-session1/VAzKDm8YHqrnjVCY5mifW1amABFKDmABFKDmB9IKDmABFKDmP6ybOm/DAOD_EGAM8.12607600._000963.pool.root.1.panda.um_010",

"Failed in data staging: Failed writing to destination: file:/disk/session/aurora-gc3-session1/9VgNDmAZHqrnjVCY5mifW1amABFKDmABFKDmaLJKDmABFKDmG9Vo5n/DAOD_EGAM8.12607600._000968.pool.root.1.panda.um_001 : Can't write to destination: No space left on device: Failed to create file /disk/session/aurora-gc3-session1/9VgNDmAZHqrnjVCY5mifW1amABFKDmABFKDmaLJKDmABFKDmG9Vo5n/DAOD_EGAM8.12607600._000968.pool.root.1.panda.um_001",

"Failed in data staging: Failed writing to destination: file:/disk/session/aurora-gc3-session1/aLBMDmWRCsrnjVCY5mifW1amABFKDmABFKDmBQjKDmABFKDmRdF4Em/HITS.11350626._012744.pool.root.1 : Can't write to destination: No space left on device: Failed to create file /disk/session/aurora-gc3-session1/aLBMDmWRCsrnjVCY5mifW1amABFKDmABFKDmBQjKDmABFKDmRdF4Em/HITS.11350626._012744.pool.root.1",

Conclusion

- Unsupervised clusterization of error log messages with Word2Vec + DBSCAN with automatic epsilon definition gives good results in quality and performance
- Next steps:
  - To analyze clusterization results in more details
  - Benchmarks on larger volumes of error messages
  - Play with tokenization, Word2Vec and DBSCAN settings for optimization